Aware of the citizenry gasping for breath, the Union government is pumping in funds worth almost Rs 6,900 crore to the National Clean Air Programme (NCAP). NCAP has 131 cities in its purview, the goal being to reduce air pollution by 20-30 per cent by next year, keeping 2017 as the base year. The task at hand is enormous. A December 2022 analysis by the NCAP tracker—a joint project by Climate Trends and Respirer Living Sciences that tracks India's clean air programme using data from continuous ambient air quality monitoring stations (CAAQMsS) across 57 Indian cities—shows worrying trends. Delhi was found to be the most polluted last year with an annual average PM2.5 concentration of 99.71 ug/m3—2.5 times the safe limit of 40 ug/m3 set by the CPCB (Central Pollution Control Board). Even among the 10 least polluted cities, the PM10 levels in nine breached the CPCB-set safe limits. Most of the least polluted cities in 2019 also saw air quality levels and rankings dip in 2022. Mumbai was one example. Since November 2022, the metropolis saw 36 days with 'poor' AQI—the highest ever in the past four years per CPCB data. It hasn't had a single day of satisfactory air quality since October 2022. The impact of such high levels of pollutants will tell in the years to come, say experts.

## THE DIAGNOSTIC CHALLENGE

"It isn't just the fact that pollution is high that is troubling," says Mumbai-based pulmonologist Dr Sarthak Rastogi. "It is also the diagnostic and treatment challenges we face with growing numbers of COPD patients. If there was more awareness of COPD, treating it on time leading to better results would be easier. Right now, we get cases at an already aggravated stage. Respiratory facilities for such patients are few, which means they have to be on medications for life. Worse still, some of them will be in such a bad condition that they will need steroids and hospitalisation."

Most patients can't recognise early signs of COPD because the symptoms mimic issues typical of old age such as tiredness and weakness. "Many patients I get tell me they are tired and don't feel like walking, or they will say they have pain in the limbs. Very few patients of early COPD come in and say they're feeling breathless though the symptom is present. It is vital for doctors to be clinically aware of other symptoms that point to a growing respiratory problem," says Delhi-based pulmonologist Dr Ashish Rohatgi.

COPD is usually diagnosed with a pulmonary function test (PFT). "The tricky part is that despite pre-COPD symptoms, the PF test will be normal. You have to be exposed for some time before you can be diagnosed with COPD. It is usually present in the 50-plus population and they often stay undiagnosed till they experience severe breathlessness or wheezing," says Dr Chawla. Sometimes, very mild symptoms such as reduced stamina in the summer months or breathlessness in the months of high pollution, typically in winters, indicate an underlying COPD problem. "Patients get surprised that they have suddenly developed a problem, but the minor signs are always there. The best advice is to consult doctors regularly, particularly if you have major risk factors—a history of smoking or long periods of residence in highly-polluted conditions," says Dr Chhajed.

Late diagnosis makes treating the condition more difficult. "You can halt the progress of the disease if it is caught on time. But by the time people come in, it is



**"IT IS VITAL FOR DOCTORS TO BE CLINICALLY AWARE OF OTHER SYMPTOMS THAT POINT TO A RESPIRATORY PROBLEM"**

**DR ASHISH ROHATGI**
Pulmonologist, Delhi

# THE ABC OF COPD

## WHAT IS COPD?



**Regular exposure** to throat irritants causes inflammation and obstruction in the airways. When the obstruction gets severe, breathing becomes difficult. This condition is known as COPD.

**According to the global** burden of disease (GBD) data in 2019, COPD was the second leading cause of death and disability-adjusted life years (DALYs) in India. The cases have increased by nearly 10 million from 1990 to 2019.

**While smoking** has traditionally been the most common cause of COPD, GBD data shows that 48 per cent of COPD deaths were due to non-smoking causes in India

**Air pollution** is the second most common cause. Several studies show that toxic air greatly increases the risk of developing COPD.

## EARLY WARNINGS FOR COPD



While breathlessness is a major symptom, doctors say there are less obvious signs preceding the advent of chronic breathlessness. These are:

- Weakness doing activities you are used to
- Pain in the limbs
- Excess mucus production
- Chest congestion
- Frequent chest infections
- Chronic persistent cough

If any of these symptoms persist, then a lung function test and a visit to a pulmonologist are advised for early detection and timely intervention for COPD.